

Shaft Design for Rotor-Pulley Assembly

Layout and Assumptions

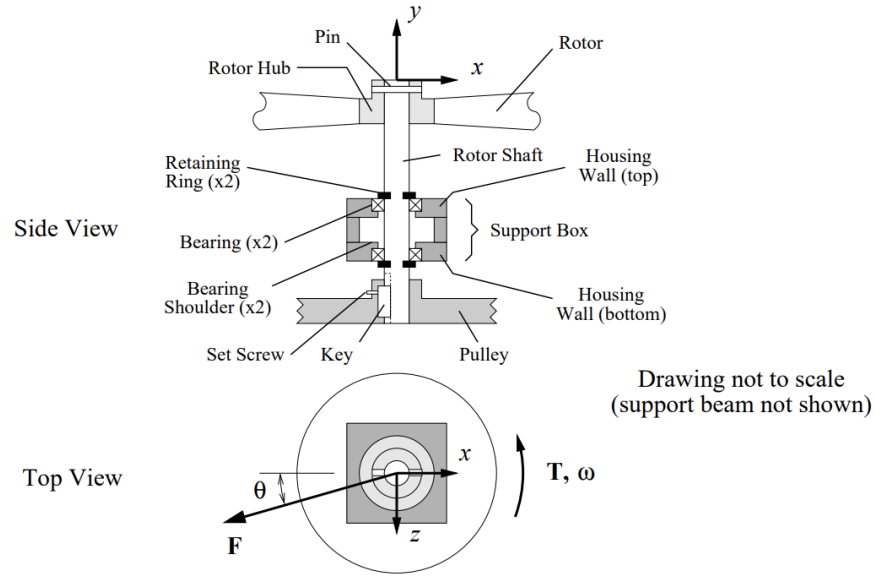


Figure 1: Assembly Layout

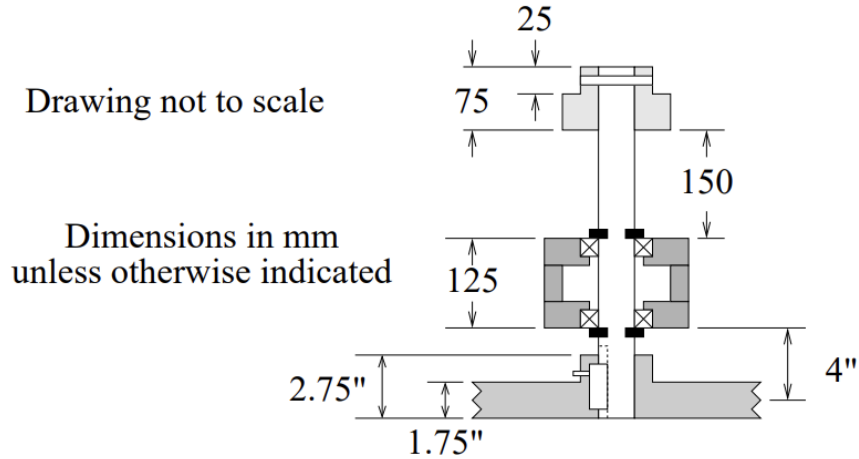


Figure 2: Assembly Dimensions

1. To allow for overloads, assume a factor of safety of 1.5 for all components.
2. Shaft must be designed for infinite life.¹
3. All components and material must be available at McMaster-Carr. Costing uses the cost/length or cost/part of the longest stocked length or quantity $\geq$ the needed dimension.
4. Material: steel or alloy steel shafts only. Assume bulk hardness $\leq$ HRC 28. Where McMaster omits strength, approximate S_{ut} from hardness.

¹Budynas & Nisbett, *Shigley's Mechanical Engineering Design*, discussion of endurance-limit design with the Goodman diagram and combined stresses (see Chs. 6–7).

5. Retaining rings should be corrosion-resistant steel; other connectors (pin, key, set screw) any suitable material.²
6. For this design only, key yield $S_y = 0.8S_{ut}$ if not published.
7. Specify shaft tolerance.
8. Do not include machining charges
9. Bearing width ≤ 0.5 in. Shaft diameter must match a standard ball bearing bore available at McMaster.
10. Assume pulley bore is any standard precision rotary shaft diameter with matching standard keyway; pulley hub dia. 2.5 in, pulley section 1.75 in, total width 2.75 in; weight ≈ 25 N on belt centerline.³
11. Rotor bore can match any standard shaft diameter. Rotor hub length 75 mm; 25 mm available for the cross pin. Hub OD at pin 65 mm. Rotor weight 30 N; support box 28 N.

From the given layout the important locations and FBDs are

$y_P = 0.875$ in	(pulley/belt/torque/hub CG)
$y_A = 4.875$ in	(bottom ring & bottom bearing)
$y_S = 9.3275$ in	(shaft CG)
$y_B = 9.796$ in	(top ring & top bearing)
$y_R = y_B + 150 \text{ mm} + \frac{75 \text{ mm}}{2} = 16.690$ in	(rotor CG)
$y_{\text{pin}} = y_B + 150 \text{ mm} + 50 \text{ mm} + \frac{25 \text{ mm}}{2} = 18.162$ in	(pin: thrust & rotor torque)
$L = 473.825 \text{ mm} = 18.655$ in	(overall length).

²Shigley notes use of manufacturer catalogs for retaining-ring load ratings and emphasizes high K_t for grooves; see Fig. 7-18 and text.

³Standard inch key sizes vs. shaft diameter per Shigley Tbl. 7-6 and Fig. 7-16.

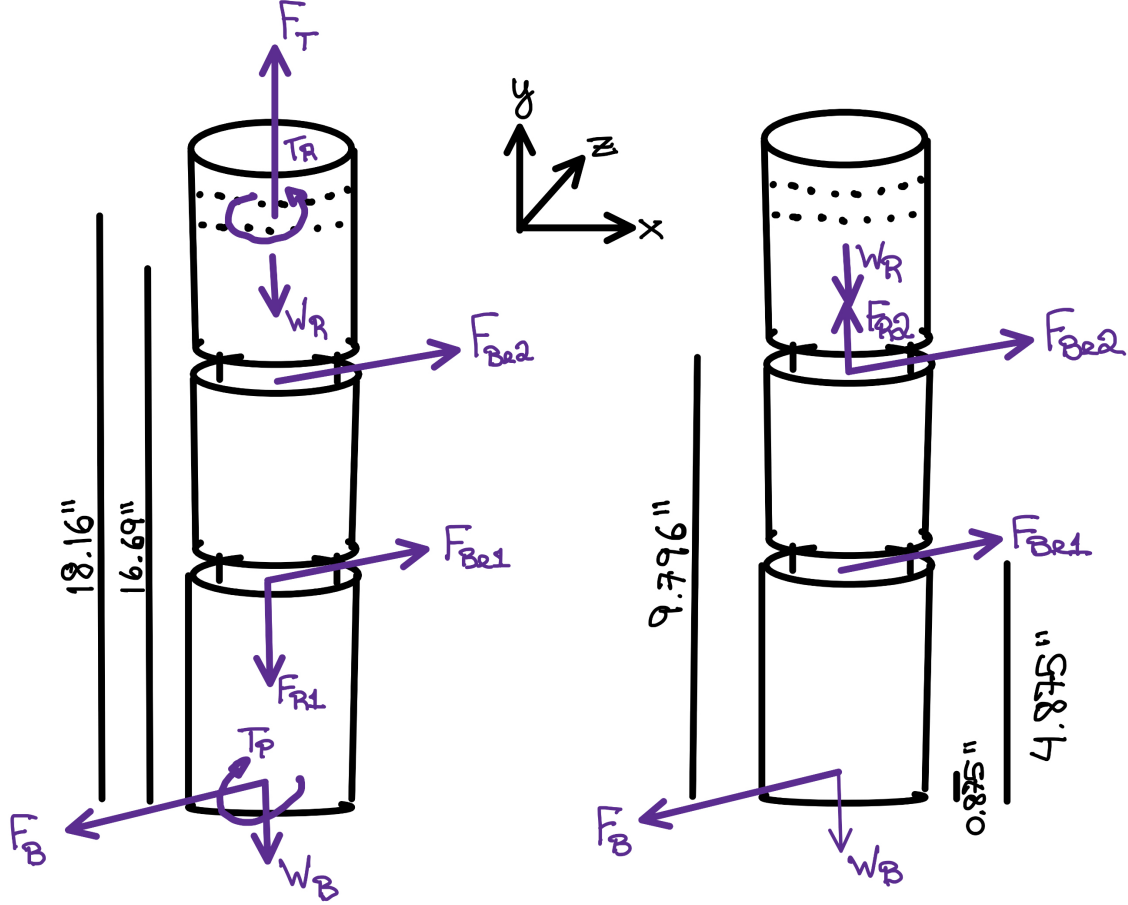


Figure 3: Free-Body Diagram of Shaft *left in operation, right shutoff*

Where F_T is the thrust force, F_{Be2} is the top bearing force, F_{Be1} is the bottom bearing force, W_R is the rotor weight, F_{R1} is the bottom ring force, W_B is the belt/pulley weight, F_B is the belt force, T_P is the pulley torque, T_R is the rotor torque.

Ring axial forces and bearing radial forces are modeled at the same location (the ring radius is small, yielding $< 3\%$ change in local moments ($1/8'' / 18.655'' \leq 1\%$)). Shaft weight is not included for these preliminary calculations as this actually causes the rings to resist higher thrust forces, being the worst-case.

Loads (Inputs to MATLAB)

Belt resultant $|\mathbf{F}| = 254.9 \text{ lbf}$ at y_P , angle $\theta = 2.329^\circ$ from $-\hat{x}$ so $F_x = -254.8 \text{ lbf}$, $F_z = 10.4 \text{ lbf}$. Torque $T = 471.89 \text{ in lbf}$ for $y \in [y_P, y_{\text{pin}}]$. Upthrust at the pin $T_y = 170.6 \text{ lbf}$. Centerline weights: pulley 5.62 lbf at y_P , rotor 6.75 lbf at y_R , and no shaft self-weight (worst case for retaining rings). Supporting box weight is not actually part of the equation, since that is held by the frame of the drone, not the supporting rings.

Statics Results and Diagrams

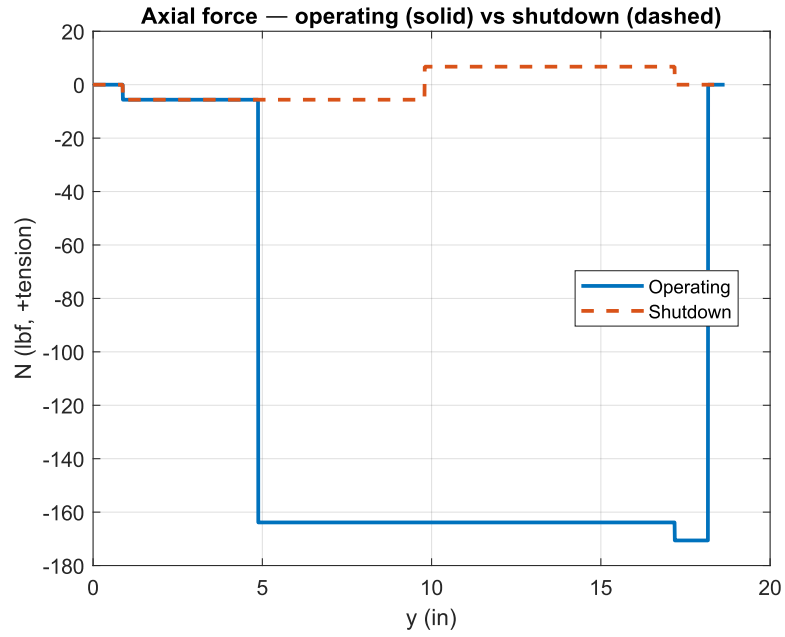


Figure 5: Axial force - drone operating (solid) vs drone off (dashed)

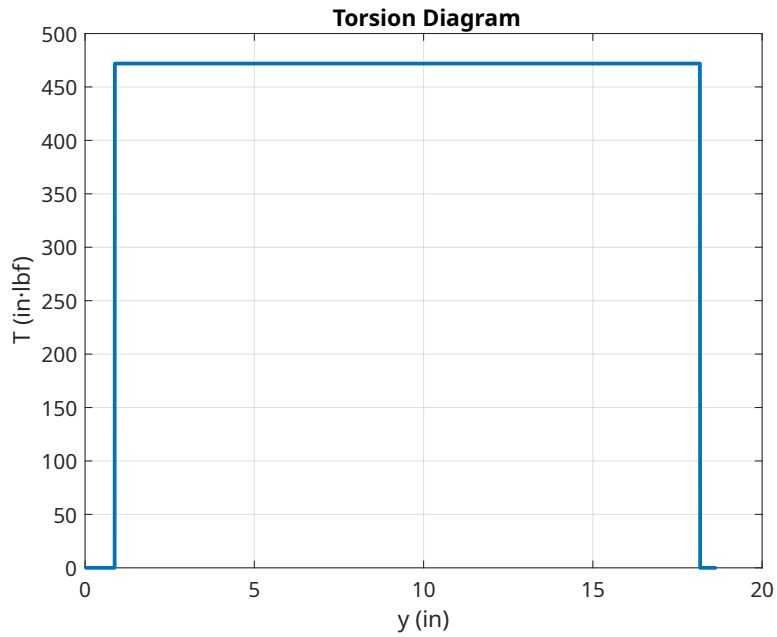


Figure 6: Torsion diagram (T is constant between pulley and pin)

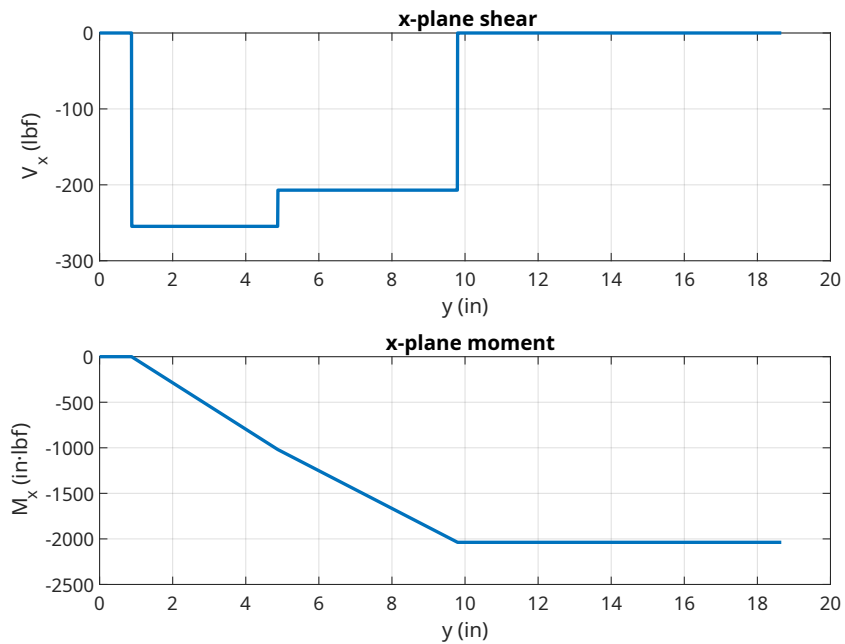


Figure 7: x -plane shear and moment

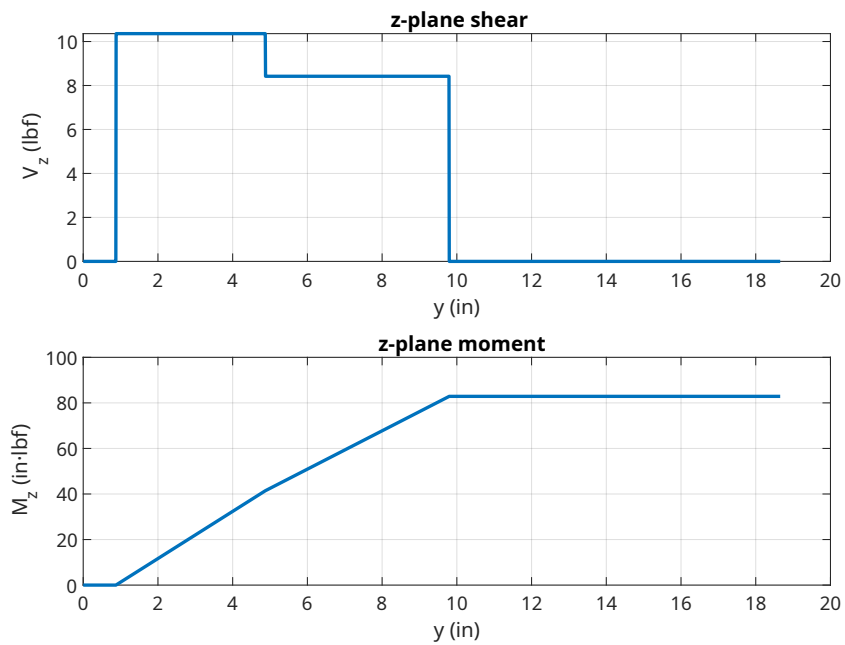


Figure 8: z -plane shear and moment

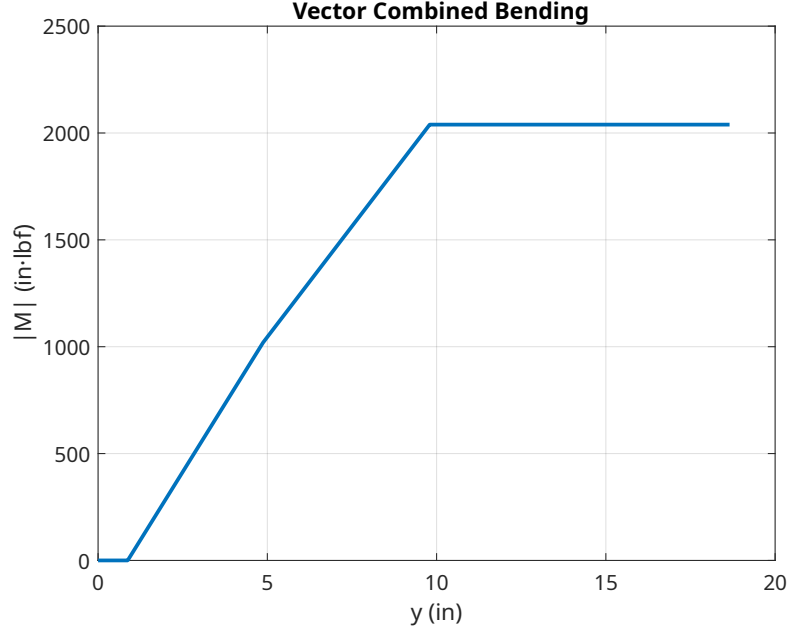


Figure 9: Vector combined bending $M(y) = \sqrt{M_x^2 + M_z^2}$

Sample statics check (bearing reactions). With $a = y_A - y_P = 4.000$ in and $L_b = y_B - y_A = 4.921$ in,

$$R_{Bx} = -F_x \frac{a}{L_b} = 207.0 \text{ lbf}, \quad R_{Ax} = -F_x - R_{Bx} = 47.7 \text{ lbf}.$$

Similarly for z : $R_{Bz} = -F_z a/L_b = -8.42 \text{ lbf}$, $R_{Az} = -1.94 \text{ lbf}$.

The vector bending moment peaks at $M_{\max} = 2039.2 \text{ in lbf}$ at $y \approx 9.9$ in and continues until the end of the beam.

For the retaining rings, when in operation, the bottom ring opposes the positive force $\Sigma F_y = 0 = -W_{\text{pulley}} + F_{\text{botring}} - W_{\text{rotor}} + F_{\text{thrust}} = -25N + F_{\text{botring}} - 28N + 758.8N \rightarrow F_{\text{botring}} \approx 713.8N \approx 160.47 \text{ lbf}$.

When not in operation, the top ring opposes the negative force $\Sigma F_y = 0 = -W_{\text{pulley}} + F_{\text{topring}} - W_{\text{rotor}} = -25N + F_{\text{topring}} - 28N \rightarrow F_{\text{topring}} = 53N \approx 11.91 \text{ lbf}$.

Fatigue (DE-Goodman) with Axial Mean Stress

The shaft sees completely reversed bending (rotating element), steady torsion in the pulley-pin segment, and a steady compressive axial force (mainly thrust) during operation. Following Shigley, we compute von Mises alternating and mean stresses (Eqs. 7-4/7-5 with axial added) and use the Goodman line (Eq. 6-41).⁴ **Alternating** terms use K_f, K_{fs} ; **mean/static** terms use K_t, K_{ts} . The sections that follow are all preliminary calculations, gaging the important variables, etc., and will all be recalculated in later sections, when the final components are decided.

⁴Budynas & Nisbett, *Shigley's Mechanical Engineering Design*, 11th ed., McGraw-Hill, 2019. Sections 6-4, 6-6, 7-3, 7-4.

Material and endurance limit

For standard cold-drawn steel, we use a preliminary choice $S_{ut} = 100$ ksi (AISI 1050 CD range).⁵ Uncorrected $S'_e = 0.5S_{ut} = 50$ ksi (for steels with $S_{ut} \leq 200$ ksi).⁶ We correct S'_e via

$$S_e = k_a k_b k_c k_d k_e S'_e.$$

Using machined/turned surface $k_a = aS_{ut}^b$ with $a = 2.00$, $b = -0.217$;⁷ size (bending) $k_b = (d/0.3)^{-0.107}$ (inches);⁸ load $k_c = 1$ (bending), temperature $k_d = 1$ (room), reliability $k_e = 0.702$ for 99.99%.⁹ This high reliability factor was chosen, since this is an application with high risk of life.

Solving for these introduces a diameter dependence (which is iterated in MATLAB).

Stress concentration and notch sensitivity

Fatigue SCFs using Peterson's model:

$$K_f = 1 + q(K_t - 1), \quad K_{fs} = 1 + q_s(K_{ts} - 1),$$

with notch sensitivity q, q_s from Shigley Fig. 6–26/6–27. For steel near $S_{ut} \approx 100$ ksi and preliminary notch radius $r = 0.020$ in we take $q \approx 0.72$, $q_s \approx 0.85$.¹⁰ For the *retaining-ring groove* the preliminary guesses are used, $K_t = 5.0$, $K_{ts} = 3.0$.¹¹ For the *pin cross-hole* we use $K_t = 3.0$, $K_{ts} = 2.6$.¹² (Keyseat is torsion-only and not governing here, but later calculated for final setup.)

Critical sections evaluated

- **Top ring/bearing groove** (just above y_B): $\max M \approx 2040 \text{ in} \cdot \text{lb}f$, T is present, axial mean ≈ -160 lbf.
Preliminary SCFs: $K_t = 5.0$, $K_{ts} = 3.0 \Rightarrow K_f \simeq 3.88$, $K_{fs} \simeq 2.7$.
- **Pin cross-hole** (just below y_{pin}): $\max M \approx 2040 \text{ in} \cdot \text{lb}f$, T is present, axial mean ≈ -170 lbf.
Preliminary SCFs: $K_t = 3.0$, $K_{ts} = 2.6 \Rightarrow K_f \simeq 2.44$, $K_{fs} \simeq 2.36$.

Stress model at a section s

With a diameter d and loads M_s, T_s, N_s ,

$$\begin{aligned} \sigma_{a,s} &= \frac{32 K_f^{(s)} M_s}{\pi d^3} \quad (\text{bending, alternating}), \\ \tau_{m,s} &= \frac{16 K_{fs}^{(s)} T_s}{\pi d^3} \quad (\text{torsion, mean}), \\ \sigma_{m,ax,s} &= K_{f,ax}^{(s)} \frac{N_s}{A}, \quad A = \frac{\pi d^2}{4} \quad (\text{axial, mean}). \end{aligned}$$

⁵“AISI 1050 Steel, cold drawn,” MatWeb Material Property Data (accessed Nov. 2025).

⁶Shigley Eq. (6–17).

⁷Shigley Table 6–2.

⁸Shigley Eq. (6–19).

⁹Shigley Table 6–5.

¹⁰Shigley Figs. 6–26 and 6–27.

¹¹Shigley Figs. A–15–14 and A–15–15; see also Ch. 7, Fig. 7–18 and text noting high K_t for grooves.

¹²Shigley Figs. A–15–10 and A–15–11.

Thus the fluctuating von Mises is:

$$\sigma'_{a,s} = \sqrt{\sigma_{a,s}^2 + 3\tau_{a,s}^2} = \sigma_{a,s}, \quad \sigma'_{m,s} = \sqrt{\sigma_{m,ax,s}^2 + 3\tau_{m,s}^2}.$$

Goodman (which is conservative) gives

$$n_s = \left(\frac{\sigma'_{a,s}}{S_e} + \frac{\sigma'_{m,s}}{S_{ut}} \right)^{-1}.$$

Example Worked Fatigue

Trial 1: $d = 1.50$ in. Size factor $k_b = (1.50/0.3)^{-0.107} = 0.842$, so

$$S_e = 0.736(0.842)(1)(1)(0.702)(50) = 21.75 \text{ ksi}.$$

Top ring groove ($K_t = 5$, $K_{ts} = 3 \Rightarrow K_f = 3.88$, $K_{fs} = 2.70$):

$$\sigma_a = \frac{32(3.88)(2039.2)}{\pi(1.50)^3} \frac{1}{1000} = 23.87 \text{ ksi},$$

$$\tau_m = \frac{16(2.70)(471.9)}{\pi(1.50)^3} \frac{1}{1000} = 2.136 \text{ ksi},$$

$$A = \frac{\pi(1.50)^2}{4} = 1.767 \text{ in}^2, \quad \sigma_{m,ax} = 5 \frac{(-160)}{1.767} \frac{1}{1000} = -0.421 \text{ ksi},$$

$$\sigma'_m = \sqrt{(-0.421)^2 + 3(2.136)^2} = 3.723 \text{ ksi},$$

$$n = \left(\frac{23.87}{21.75} + \frac{3.723}{100} \right)^{-1} = \mathbf{0.88} \quad (\text{fails } n = 1.5).$$

Pin cross-hole ($K_t = 3$, $K_{ts} = 2.6 \Rightarrow K_f = 2.44$, $K_{fs} = 2.36$):

$$\sigma_a = \frac{32(2.44)(2039.2)}{\pi(1.50)^3} \frac{1}{1000} = 15.02 \text{ ksi},$$

$$\tau_m = \frac{16(2.36)(471.9)}{\pi(1.50)^3} \frac{1}{1000} = 1.679 \text{ ksi},$$

$$\sigma_{m,ax} = 3 \frac{(-170.6)}{1.767} \frac{1}{1000} = -0.290 \text{ ksi},$$

$$\sigma'_m = \sqrt{(-0.290)^2 + 3(1.679)^2} = 2.923 \text{ ksi},$$

$$n = \left(\frac{15.02}{21.75} + \frac{2.923}{100} \right)^{-1} = \mathbf{1.39} \quad (\text{fails}).$$

Ring-governing diameter. Because $\sigma_a, \tau_m \propto d^{-3}$ and S_e varies somewhat little for k_b , solving

$$\frac{\sigma'_a}{S_e} + \frac{\sigma'_m}{S_{ut}} = \frac{1}{1.5} \text{ for the ring groove gives}$$

$$d_{\text{ring}} \approx \boxed{1.82 \text{ in}}.$$

So we check at $d = 1.82$ in (ring):

$$k_b = (1.82/0.3)^{-0.107} = 0.825, \quad S_e = 0.736(0.825)(0.702)(50) = 21.30 \text{ ksi},$$

$$\sigma_a = \frac{32(3.88)(2039.2)}{\pi(1.82)^3} \frac{1}{1000} = 13.36 \text{ ksi}, \quad \tau_m = \frac{16(3.00)(471.9)}{\pi(1.82)^3} \frac{1}{1000} = 1.198 \text{ ksi},$$

$$A = \frac{\pi(1.82)^2}{4} = 2.600 \text{ in}^2, \quad \sigma_{m,ax} = 5 \frac{(-160)}{2.600} \frac{1}{1000} = -0.289 \text{ ksi},$$

$$\sigma'_m = \sqrt{(-0.289)^2 + 3(1.198)^2} = 2.096 \text{ ksi}, \quad n = \left(\frac{13.36}{21.30} + \frac{2.096}{100} \right)^{-1} = \mathbf{1.54} \text{ (OK)}.$$

Pin check.

$$\begin{aligned}\sigma_a &= \frac{32(2.44)(2039.2)}{\pi(1.82)^3} \frac{1}{1000} = 8.40 \text{ ksi}, \quad \tau_m = \frac{16(2.36)(471.9)}{\pi(1.82)^3} \frac{1}{1000} = 0.942 \text{ ksi}, \\ \sigma_{m,\text{ax}} &= 3 \frac{(-170.6)}{2.600} \frac{1}{1000} = -0.197 \text{ ksi}, \quad \sigma'_m = \sqrt{(-0.197)^2 + 3(0.942)^2} = 1.643 \text{ ksi}, \\ n &= \left(\frac{8.40}{21.30} + \frac{1.643}{100} \right)^{-1} = \mathbf{2.44} \text{ (OK)}.\end{aligned}$$

With factors $(K_t, K_{ts}) = (5, 3)$, the **ring groove governs**.

The nearest standard size would be $d = 1.875 \text{ in}$. For this size, $k_b = (1.875/0.3)^{-0.107} = 0.822$ and $S_e = 0.736(0.822)(0.702)(50) = 21.2 \text{ ksi}$.

Ring groove @ $d = 1.875 \text{ in}$:

$$\begin{aligned}\sigma_a &= 12.22 \text{ ksi}, \quad \tau_m = 1.094 \text{ ksi}, \quad A = 2.764 \text{ in}^2, \quad \sigma_{m,\text{ax}} = -0.259 \text{ ksi}, \\ \sigma'_m &= 1.914 \text{ ksi}, \quad n = \left(\frac{12.22}{21.2} + \frac{1.914}{100} \right)^{-1} = \mathbf{1.68} \text{ (OK)}.\end{aligned}$$

Pin hole @ $d = 1.875 \text{ in}$:

$$\begin{aligned}\sigma_a &= 7.689 \text{ ksi}, \quad \tau_m = 0.862 \text{ ksi}, \quad \sigma_{m,\text{ax}} = -0.185 \text{ ksi}, \\ \sigma'_m &= 1.504 \text{ ksi}, \quad n = \left(\frac{7.689}{21.2} + \frac{1.504}{100} \right)^{-1} = \mathbf{2.65} \text{ (OK)}.\end{aligned}$$

Yield check with $d = 1.875 \text{ in}$ (static, so we use K_t, K_{ts})

For this, we use von Mises maximum $\sigma_{\text{VM,max}} = \sqrt{(\sigma_{b,\text{max}} + \sigma_{m,\text{ax}})^2 + 3\tau_m^2}$ with

$$\sigma_{b,\text{max}} = \frac{32K_t M}{\pi d^3}, \quad \tau_m = \frac{16K_{ts} T}{\pi d^3}, \quad \sigma_{m,\text{ax}} = K_t \frac{N}{A}.$$

Take $S_y = 100 \text{ ksi}$ (1045 HRC28, $d > 0.75 \text{ in}$, per Piazza).¹³

- **Top ring groove** ($M=2039.2$, $N \approx -160$, $K_t=5$, $K_{ts}=3$):
 $\sigma_{b,\text{max}} = 15.76 \text{ ksi}$, $\tau_m = 1.095 \text{ ksi}$, $\sigma_{m,\text{ax}} = -0.259 \text{ ksi}$
 $\Rightarrow \sigma_{\text{VM,max}} = 15.62 \text{ ksi}$, $n_y = \frac{S_y}{\sigma_{\text{VM,max}}} = \mathbf{6.4}$.
- **Bottom ring groove** ($M \simeq 1.02 \times 10^3 \text{ in lbf}$; same N):
 $\sigma_{b,\text{max}} = 7.886 \text{ ksi}$, $\tau_m = 1.095 \text{ ksi}$, $\sigma_{m,\text{ax}} = -0.262 \text{ ksi}$
 $\Rightarrow \sigma_{\text{VM,max}} = 7.86 \text{ ksi}$, $n_y = \mathbf{12.7}$.
- **Pin hole** ($M=2039.2$, $N \approx -170.6$, $K_t=3$, $K_{ts}=2.6$):
 $\sigma_{b,\text{max}} = 9.456 \text{ ksi}$, $\tau_m = 0.949 \text{ ksi}$, $\sigma_{m,\text{ax}} = -0.185 \text{ ksi}$
 $\Rightarrow \sigma_{\text{VM,max}} = 9.42 \text{ ksi}$, $n_y = \mathbf{10.6}$.
- **Keyseat at pulley** (torsion is a lot greater than bending; as Shigley suggests, take $K_{ts} \approx 3$):
 $\tau_m = \frac{16(3)T}{\pi d^3} = 1.095 \text{ ksi}$, $\sigma_{\text{VM,max}} \approx \sqrt{3} \tau_m = 1.90 \text{ ksi}$
 $\Rightarrow n_y = \mathbf{52.7}$ (very large).

All locations satisfy static yielding easily with $d = 1.875 \text{ in}$.

¹³Alpha. "Yield Strength Values for McMaster-Carr Rotary Shaft Materials." Piazza, course DP2 note, posted Nov. 2025.

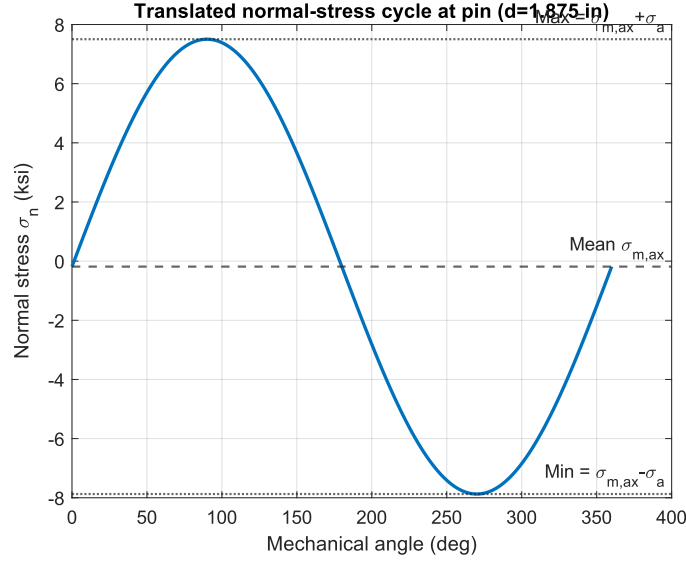


Figure 10: Stress plot over one revolution at the pin section ($d = 1.875$ in). The bending term is sinusoidal (alternating), the axial and torsion terms are mean (steady).

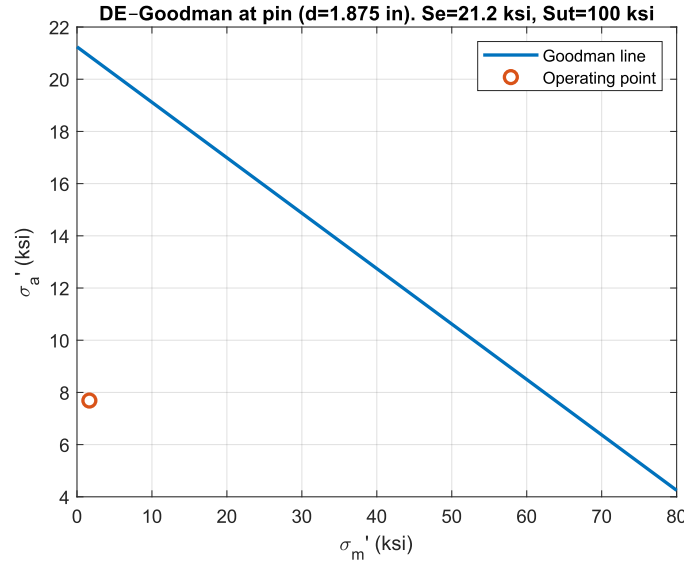


Figure 11: Goodman diagram at the pin section showing σ'_a vs. σ'_m and S_e line (DE-Goodman). The operating point for $d = 1.875$ in lies below the line (safe).

Deflection and Slope Check (stiffness)

For this, we treat the shaft as a beam with uniform stiffness EI (the local grooves/keyways/etc. do not control stiffness). With $E = 30$ Msi and $I = \pi d^4/64$, the curvature is

$$\kappa(y) = \frac{M(y)}{EI}, \quad \theta'(y) = \kappa(y), \quad v'(y) = \theta(y).$$

We integrate numerically with the cumtrapz function in Matlab in each principal plane and simple supports at the bearings: $v(y_A) = v(y_B) = 0$.¹⁴ With $d = 1.875$ in, the bearing slopes are

¹⁴Shigley Sec. 7-2; also see Example 7-3 for slope/deflection modeling of shafts.

small: $|\theta_A| \approx 12 \times 10^{-4}$ rad, $|\theta_B| \approx 3.0 \times 10^{-4}$ rad. (Plots included.), which are well below typical slope limits for ball bearings and even cylindrical bearings¹⁵.

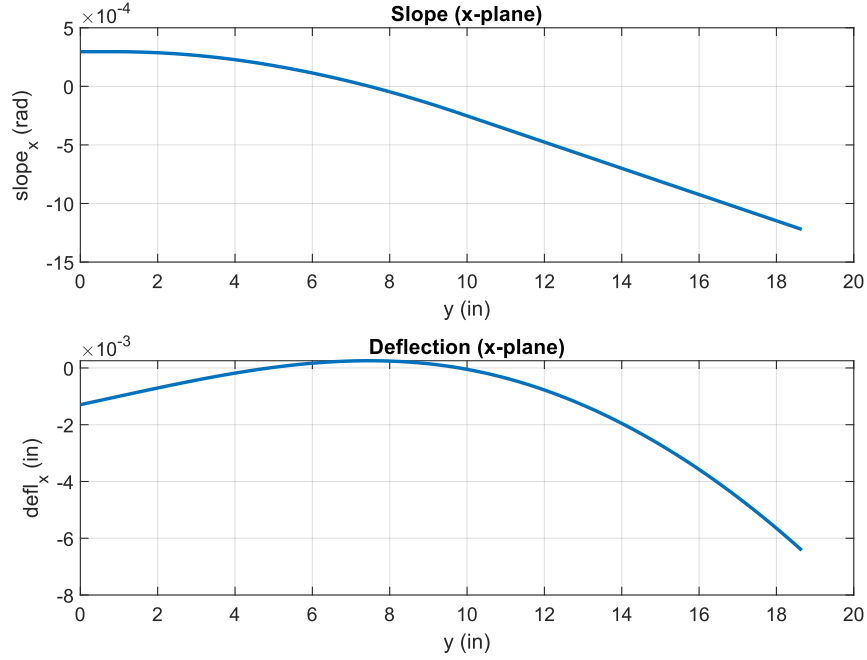


Figure 13: Slope and deflection in the x -plane ($d = 1.875$ in)

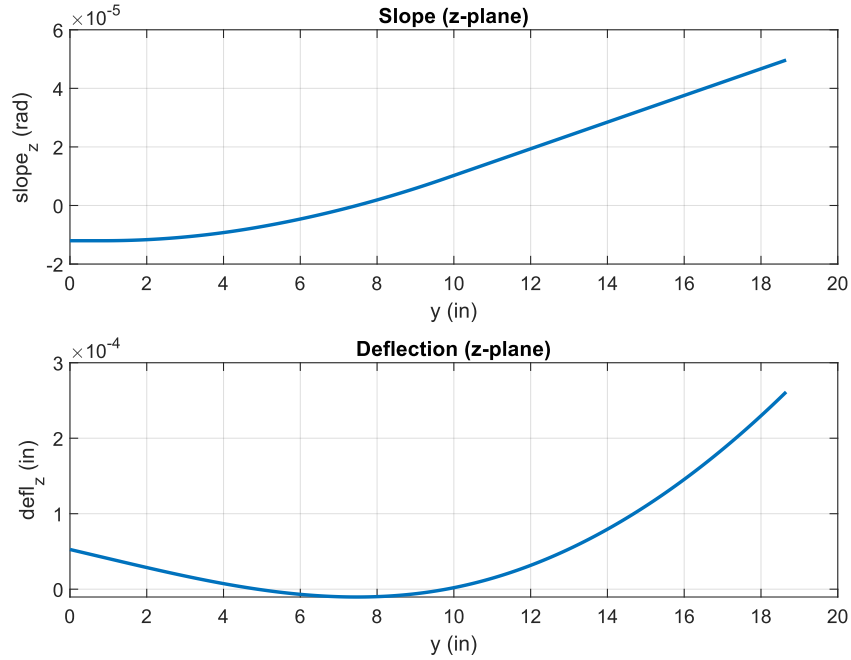


Figure 14: Slope and deflection in the z -plane ($d = 1.875$ in)

¹⁵See Shigley Tbl. 7-2 (typical bearing slope considerations) and surrounding text.

Pulley Connection Hardware: Rectangular Key and Set Screw

A rectangular key transmits the pulley torque into the shaft. Shigley Table 7–6 gives standard inch key sizes vs. shaft diameter (Fig. 7–16a)¹⁶ For our selected shaft $d = 1.875$ in, the table range 1.75–2.25 in maps to a **1/2 in square key** ($w = h = 0.500$ in).¹⁷ The pulley hub width is 2.75 in, and Shigley recommends a maximum key length of about $1.5d = 2.8125$ in to avoid twist issues¹⁸, so the full hub width $\ell = 2.75$ in is acceptable.

Forces at the key

The torque from the belts is $T = 471.89$ in lbf (from statics). The tangential force at the shaft surface is

$$F = \frac{T}{r} = \frac{471.89}{1.875/2} = 503.35 \text{ lbf.}$$

This is the force that tends to shear the key and crush its face (Shigley Fig. 7–19 and Ex. 7–6).¹⁹

Key strength checks (shear and bearing)

Following Shigley Ex. 7–6, we check:

$$\begin{aligned} \text{Shear: } \tau &= \frac{F}{w \ell} \leq \frac{S_{sy}}{n}, & S_{sy} &= 0.577 S_y \quad (\text{DE theory}), \\ \text{Bearing: } \sigma &= \frac{F}{(h \ell / 2)} \leq \frac{S_y}{n}. \end{aligned}$$

Using a key stock cold-draw low-carbon steel ($S_y = 54$ ksi)²⁰, with the project design factor $n = 1.5$ and the standard key size $w = h = 0.500$ in:

$$\ell_{\text{req, shear}} = \frac{n F}{S_{sy} w} = \frac{1.5 \cdot 503.35}{(0.577)(54,000) \cdot 0.5} = \boxed{0.048 \text{ in}},$$

$$\ell_{\text{req, bearing}} = \frac{2n F}{S_y h} = \frac{2 \cdot 1.5 \cdot 503.35}{54,000 \cdot 0.5} = \boxed{0.056 \text{ in}}.$$

Both required lengths are a lot smaller than the available hub width, so it passes. Still, using the real dimensions $\ell = 2.75$ in, the actual stresses are

$$\tau = \frac{F}{w \ell} = \frac{503.35}{0.5 \cdot 2.75} = 0.365 \text{ ksi}, \quad \sigma = \frac{F}{(h \ell / 2)} = \frac{503.35}{0.5 \cdot 2.75 / 2} = 0.732 \text{ ksi.}$$

So $n_{\text{shear}} = S_{sy} / \tau \approx \mathbf{85}$ and $n_{\text{bearing}} = S_y / \sigma \approx \mathbf{74}$. The key easily meets the stress requirements with the selected size and length.

¹⁶Shigley, Tbl. 7–6; Fig. 7–16 (standard inch keys and selection guidance).

¹⁷Shigley, Tbl. 7–6 and discussion above it (stock, lengths, and length limit $\leq 1.5d$).

¹⁸Shigley, Ex. 7–6 and associated text.

¹⁹Shigley Fig. 7–19; Ex. 7–6 (key shear and bearing checks with DE shear $S_{sy} = 0.577 S_y$).

²⁰Shigley, Ex. 7–6 (uses UNS G10180 with $S_y \approx 54$ ksi as typical key stock).

Set screw for axial

Because the belts act in the pulley plane, axial load on the pulley is small; it must be able to just keep the pulley from drifting axially on the shaft at equilibrium. For that, we can use a radial set screw pressing on the *top of the key*. The axial force comes from friction at two surfaces (key-hub and key-shaft). With a conservative steel-on-steel coefficient $\mu = 0.15$,

$$F_{\text{ax,cap}} = \underbrace{2\mu}_{\text{two interfaces}} N,$$

where N is the radial normal created by the set screw.²¹

Axial force to hold. Taking the pulley weight as the axial load and including the factor of safety:

$$F_{\text{ax,req}} = n W_{\text{pulley}} = 1.5 \times 5.62 \approx 8.43 \text{ lbf.}$$

The required normal force is then

$$N_{\text{req}} = \frac{F_{\text{ax,req}}}{2\mu} = \frac{8.43}{0.30} = \boxed{28.1 \text{ lbf}}.$$

Local bearing on the key under the screw tip Using a flat point set screw, we can limit the compressive (bearing) stress on the key top to $\leq S_y/n$. Approximating the projected contact with the screw's root diameter d_{root} ,

$$p = \frac{N_{\text{req}}}{\frac{\pi}{4} d_{\text{root}}^2} \leq \frac{S_y}{n} = \frac{54 \text{ ksi}}{1.5} = 36 \text{ ksi.}$$

So a #10-32 set screw ($d_{\text{root}} \approx 0.159 \text{ in}$) gives $p \approx 1.42 \text{ ksi}$ ($\ll 36 \text{ ksi}$); a 1/4-20 ($d_{\text{root}} \approx 0.188 \text{ in}$) gives $p \approx 1.01 \text{ ksi}$. Either is acceptable with very large margin and will later be chosen with the full setup.²²

Thread engagement in the pulley wall. With hub $D_{\text{hub}} = 2.5 \text{ in}$ and bore $d = 1.875 \text{ in}$, the thickness is

$$t_{\text{hub}} = \frac{D_{\text{hub}} - d}{2} = 0.3125 \text{ in.}$$

The recommended engagement for steel is $\gtrsim 1 \times d_{\text{root}}$; both #10-32 and 1/4-20 meet this ($t_{\text{hub}}/d_{\text{root}} \approx 1.64$ and 1.66 , respectively).

Rotor Connection

For this, we need to size a single-shear cross pin that transmits the rotor torque without exceeding pin shear, hub bearing (crushing), hub tear-out, and contact given the same minimum edge distance used in the other design problem (0.5"). The rotor hub yield is 50 ksi; design factor $n = 1.5$. We can use von Mises shear yield for ductile materials $S_{sy} \approx 0.577 S_y$.²³

²¹Set-screw styles and typical holding powers per size: Shigley Fig. 7-15 and Tbl. 7-4. Also, Shigley notes setscrew length $\approx \frac{1}{2}d$ and hub radial thickness practice (p. 401).

²²Representative #10-32 cup-point 316 SS set screw page lists size and hardness; see McMaster part 92313A832.

²³Distortion-energy shear yield relation used consistently in Shigley (e.g., Ex. 7-6).

Loads For a shaft diameter be d and radius $r = d/2$, the pin sees shear from transmitted torque T :

$$F_T = \frac{T}{r}, \quad (1)$$

and any additional shear F_{ext} (from external forces at the rotor). We use the maximum between F_T and F_{ext} for V .

Pin shear For $n_s = 1$ shear planes:

$$\tau_{\text{pin}} = \frac{V}{n_s A} \leq \frac{0.577 S_{y,\text{pin}}}{n}, \quad A = \frac{\pi d_p^2}{4} \quad (2)$$

So the required diameter for shear is

$$d_{p,\text{shear}} = \sqrt{\frac{4V}{\pi n_s (0.577 S_{y,\text{pin}}/n)}} \quad (3)$$

Crushing With the effective bearing thickness t_{hub} :

$$p_{\text{hub}} = \frac{V}{t_{\text{hub}} d_p} \leq \frac{S_{y,\text{hub}}}{n} \Rightarrow d_{p,\text{bear}} = \frac{V}{t_{\text{hub}} (S_{y,\text{hub}}/n)} \quad (4)$$

Hub tear-out & edge distance For a single-lug:

$$V \leq 2 e_{\min} t_{\text{hub}} \left(\frac{0.577 S_{y,\text{hub}}}{n} \right), \quad (5)$$

where $e_{\min}$ is the same minimum edge distance of 0.5". We will use $e \gtrsim 1.5 d_p$ and $e \geq 0.5 \text{ in.}$ If we connect the rotor hub to the $d = 1.875 \text{ in}$ shaft with a tapered cross pin (1:48 taper, ASME B18.8), with torque at the rotor end $T = 471.89 \text{ in lbf}$ (same as the pulley), it gives the pin shear load

$$F = \frac{T}{r} = \frac{471.89}{0.9375} = 503.35 \text{ lbf.}$$

The hub OD is $D_{\text{hub}} = 2.559 \text{ in} \Rightarrow$ radial wall $t_{\text{wall}} = (2.559 - 1.875)/2 = 0.342 \text{ in.}$ Using a **taper pin, size 2/0** (large-end diameter $D_{\text{large}} = 0.1423 \text{ in}$), hardened carbon/alloy steel ($S_y \approx 120 \text{ ksi}$, through hardness), with length $L \approx 2.5 \text{ in}$ (across the hub), the diameters at the critical cross-sections (distance measured from the large end) are

$$\begin{aligned} d_1 &= D_{\text{large}} - 0.0208 x \Big|_{x=t_{\text{wall}}} = 0.1423 - 0.0208(0.3125) = 0.1358 \text{ in,} \\ d_2 &= D_{\text{large}} - 0.0208 x \Big|_{x=t_{\text{wall}}+d/2+d/2} = 0.0968 \text{ in,} \\ d_3 &= D_{\text{large}} - 0.0208 x \Big|_{x=t_{\text{wall}}+d+t_{\text{wall}}} = 0.0903 \text{ in.} \end{aligned}$$

²⁴Taper-pin geometry $d = D - 0.0208L$ per Shigley Eq. (7-35); see also ASME B18.8.2 as noted on typical vendor pages.

Pin shear

$$\tau_{\text{pin}} = \frac{F}{\frac{\pi}{4}(d_1^2 + d_2^2)} = \frac{503.35}{\frac{\pi}{4}(0.1358^2 + 0.0968^2)} = 23.0 \text{ ksi.}$$

The allowable shear (von Mises) with $n = 1.5$ is
 $0.577 S_y/n = 0.577(120)/1.5 = 46.2 \text{ ksi} \Rightarrow n_\tau = \mathbf{2.00}.$

Bearing (crushing) Each wall takes $F/2$ in bearing:

$$p_{\text{hub},1} = \frac{F/2}{t_{\text{wall}} d_1} = 5.93 \text{ ksi}, \quad p_{\text{hub},3} = \frac{F/2}{t_{\text{wall}} d_3} = 8.92 \text{ ksi.}$$

With hub $S_y = 50 \text{ ksi}$ and $n = 1.5$, allowable bearing is $33.3 \text{ ksi} \Rightarrow n_{\text{brg,hub}} = \{5.6, 3.7\}$. Bearing in the shaft (using the mid-shaft diameter $d_{\text{mid}} = 0.1163 \text{ in}$) is

$$p_{\text{shaft}} = \frac{F}{d d_{\text{mid}}} = \frac{503.35}{1.875 \cdot 0.1163} = 2.31 \text{ ksi},$$

far below 50 ksi allowable (FoS > 20).

Tear-out Single-lug tear-out capacity

$$V_{\text{to}} = 2 e t_{\text{wall}} \left(\frac{0.577 S_y}{n} \right) = 2817 \text{ lbf } (\gg 503.35 \text{ lbf}),$$

so $n_{\text{tear}} = \mathbf{5.6}.$

Taper-fit contact (press/interference) check

Because the taper pin is driven into a reamed 1:48 hole, a small diametral interference δ_d produces a radial contact pressure p on the hub wall (worst near the large end). Approximating the local section as a press fit of a solid shaft (pin) of radius a into a thick cylinder (hub) of outer radius b , Shigley's press/shrink-fit model gives

$$p = \frac{\delta_d}{a} \left[\underbrace{\frac{1 - \nu_p^2}{E_p}}_{\text{pin}} + \underbrace{\frac{1 - \nu_h^2}{E_h} \frac{b^2 + a^2}{b^2 - a^2}}_{\text{hub}} \right]^{-1}, \quad \sigma_{\theta,i} = p \frac{b^2 + a^2}{b^2 - a^2},$$

where $\sigma_{\theta,i}$ is the hub's tangential (hoop) stress at the bore. This is later checked against the real geometry, as it changes significantly with diameter, so doing preliminary calculations is not valuable.²⁵

External Retaining Ring

By the forces calculated before, we need a **corrosion-resistant retaining ring** for a 1.875 in shaft. Axial thrust to retain (with design factor) is

$$F_{\text{req}} = n F_{\text{axial}} = 1.5 \times 150 = 225 \text{ lbf.}$$

²⁵Press/shrink-fit thick-cylinder model as presented in Shigley (interference fits; Lamé solution application).

From researching, rings at this size have thrust capacities far *exceeding* this (thousands of lbf)²⁶; still, we can average pressure (for a spring steel SAE 1074-1095):

$$\begin{aligned} W_g &= 0.082(2) \text{ in } \quad (\text{groove width}), \\ D_g &= 1.805(3) \text{ in } \quad (\text{bottom diameter}), \\ t_{\text{depth}} &= (d - D_g)/2 = 0.035 \text{ in}, \quad r_{\text{bottom}} \approx 0.1 W_g \approx 0.008 \text{ in}. \end{aligned}$$

Average groove wall bearing pressure:

$$p_{\text{avg}} = \frac{F_{\text{req}}}{W_g \pi D_g} = \frac{225}{0.082 \cdot \pi \cdot 1.805} \approx 0.48 \text{ ksi } (\ll 33.3 \text{ ksi allowable}).$$

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Validation with Real McMaster Components

Based on all of the preceding preliminary calculations, the real McMaster components were chosen, attempting to match the geometry and materials as closely as possible, as well as iterating through cost. The final components' selection is below.

Selected parts and properties

Item	Choice	Notes (McMaster page data)
Shaft	Partially Keyed Rotary Shaft, 1045 steel, $d = 1.750$ in, keyway $3/8 \times 3/16$. Part Number: 6117K208 ²⁸	Turned, precision ground & polished; Brinell ≈ 190 ; diameter tol. -0.002 to 0 in; straightness ≤ 0.003 in/ft; length tol. 0 to 0.125 in.
Key	Rectangular (parallel) key, $w = h = 0.375$ in, length $\ell \leq 1.5d = 2.625$ in	Comes with the shaft and matches $3/8$ keyway listed for the it (tolerance-wise as well).
Set screw	Cup-point, 316 SS, #10-32, $3/4$ ". Part Number: 92313A832 ²⁹	Rockwell B80 (from research, $S_y \approx 35$ ksi, $S_{ut} \approx 85$ ksi) ³⁰ .
Pin	18-8 Stainless Steel <i>taper pin</i> , size 0, large end = 0.156 in, small end = 0.094 in, 1:48 taper, 2 in long. Part Number: 90681A169 ³¹	McMaster lists "Breaking strength" (single-shear) ≈ 54 ksi; ASME B18.8.2; large-end dia tolerance -0.0007 to $+0.0013$ in.
Retaining ring	Single-turn spiral internal ring for 1.750 in shaft, 1060-1090 spring steel. Part Number: 91663A640 ³²	Thrust load capacity ≈ 7160 lbf; ring thickness ≈ 0.031 in; groove width $W \approx 0.039$ in; <i>for groove</i> diameter ≈ 1.807 in.

²⁶see part 91663A640.

²⁷For typical spiral internal rings near 1.75 in shaft size, McMaster lists a thrust load capacity of order 7,160 lbf and groove geometry near $W \approx 0.039$ in, $D_g \approx 1.807$ in; see part 91663A640.

Material strengths used. For cold-drawn 1045 at BHN $\approx$ 190: we use $S_y = 83$ ksi, $S_{ut} = 97$ ksi (Piazza's DP2 Answer).³³ For the key stock (steel), per assignment: $S_y = 0.8S_{ut}$ if not published. For the rotor hub, $S_y = 50$ ksi. For spring-steel rings, the manufacturer thrust rating is given.³⁴

Geometry that changes the loads

Real shaft diameter (1.750") adds the shaft self-weight; with $L = 18.655$ in and $\rho \approx 0.283$ lb/in³,

$$W_{\text{shaft}} = \rho \frac{\pi d^2}{4} L = 12.68 \text{ lbf.}$$

So axial ring loads are:

$$\text{Operating (bottom ring)} : F_{\text{down}} = 170.6 - (5.62 + 6.75 + 12.68) = \boxed{145.6 \text{ lbf}},$$

$$\text{Shutdown (top ring)} : F_{\text{up}} = \boxed{25.1 \text{ lbf}}.$$

Radial bearing reactions from the belts are unchanged by d : $R_A = (47.68, -1.94)$ lbf, $R_B = (207.01, -8.42)$ lbf, with $|M|_{\text{max}} = 2039$ in $\cdot$ lbf for $y \in [9.9 \text{ in, end}]$, and $T = 471.89$ in $\cdot$ lbf from the pulley to the rotor pin.

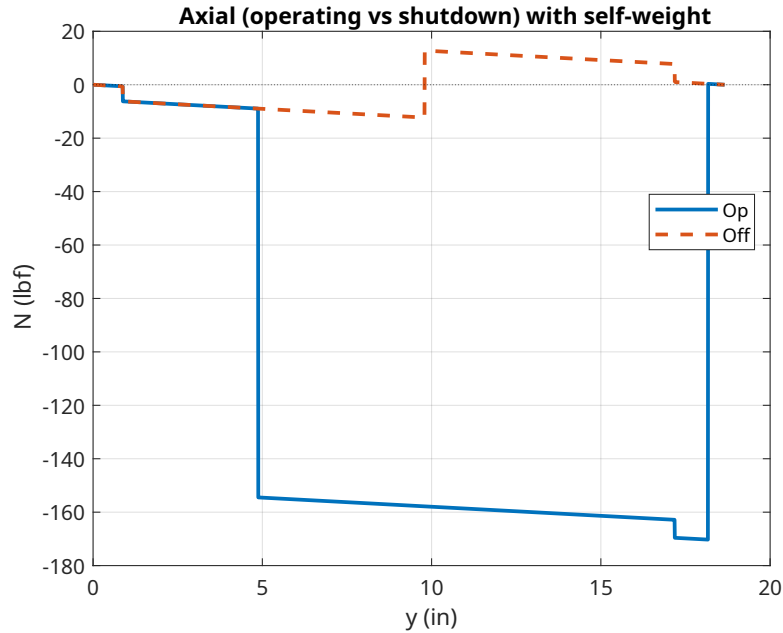


Figure 15: Axial force - drone operating (solid) vs drone off (dashed) with self-weight

Fatigue Analysis for the *actual* features (with tolerances)

- **Retaining-ring grooves (bending/torsion/axial):** from Shigley flat-bottom groove charts. Groove width $a = W = 0.039$ in; conservative groove depth $t = 0.059$ in (embed $\geq$ half ring; this is a design choice); to keep K_t and K_{ts} around same values, bottom radius $r \approx 0.2t \Rightarrow 0.012$ in should be machined. This gives $a/t \approx 0.66$ and $r/t \approx 0.2 \Rightarrow K_t = \boxed{4.75}$, $K_{ts} = \boxed{3.40}$. *Uncertainty:* using vendor tolerances $W = 0.039 \pm 0.003$ in

³³Alpha. "Yield Strength Values for McMaster-Carr Rotary Shaft Materials." Piazza, course DP2 note, posted Nov. 2025.

³⁴See retaining-ring product page referenced above.

and $D_g = 1.807 \pm 0.005 \text{ in}$ moves a/t between ~ 0.61 - 0.79 ; designing for the *worst* 0.61 ($K_t \uparrow \approx 4.76$, $K_{ts} \uparrow \approx 3.42$).³⁵

- **Rotor cross-hole at pin:** to use this pin, we must align its center length to the center of the shaft, meaning 0.125 in will protrude to both sides of the rotor hub. This means, the hole sizes should be:

$$d_{\text{hole-large}} = d_{\text{pin-small}} + (d_{\text{pin-large}} - d_{\text{pin-small}}/l_{\text{pin}} * (D_{\text{pinlarge}} - 0.125) \approx 0.153 \text{ in}$$

$$d_{\text{hole-small}} = d_{\text{pin-small}} + (d_{\text{pin-large}} - d_{\text{pin-small}}/l_{\text{pin}} * (0.125) \approx 0.117 \text{ in}$$

So using the transverse hole charts from Shigley's with worst-case ratio (because of pin tolerance) $d/D = 0.15553/1.750 \approx 0.089 \Rightarrow K_t = \boxed{2.23}$, $K_{ts} = \boxed{2.92}$ (conservative: since we are using the large diameter and assuming same diameter through all, K_t and K_{ts} are actually worse than what a tapered cut would have).³⁶

- **Pulley keyseat:** stress concentration tables could not be found for a keyseat on a shaft, so we are using Shigley's recommendation of $r/d \approx 0.02$, which gives $K_t \approx \boxed{2.14}$ (bending), $K_{ts} \approx \boxed{3.00}$ (torsion, key in place).³⁷
- **Notch sensitivity (steel, $S_{ut} \approx 97 \text{ ksi}$):** as per Shigley's Figure 6-26 6-27, with $r=0.05 \text{ in}$ use $q \approx 0.75$ (bending), $q_s \approx 0.85$ (torsion) so $\Rightarrow K_f = 1 + q(K_t - 1) = \boxed{1.92}$, $K_{fs} = 1 + q_s(K_{ts} - 1) = \boxed{2.63}$.³⁸

Endurance limit with real finish and size

Machined/Turned surface: $k_a = aS_{ut}^b$ with $(a, b) = (2.00, -0.217)$, $k_a = 0.75$. Size (bending) $k_b = (d/0.3)^{-0.107}$ with $d = 1.750 \text{ in} \Rightarrow k_b = 0.828$. $k_c = 1$ (bending), $k_d = 1$ (room), reliability $k_e = 0.702$ (99.99%). With $S'_e = 0.5S_{ut} = 48.5 \text{ ksi}$,

$$S_e = k_a k_b k_c k_d k_e S'_e = (0.75)(0.828)(1)(1)(0.702)(48.5) = \boxed{28.2 \text{ ksi}}.$$

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Fatigue (DE-Goodman) at critical sections

Just like before, we will use alternating bending with K_f , mean torsion with K_{fs} , and mean axial with K_t :

$$\sigma_a = \frac{32K_f M}{\pi d^3}, \quad \tau_m = \frac{16K_{fs} T}{\pi d^3}, \quad \sigma_{m,ax} = K_t \frac{N}{A}, \quad A = \frac{\pi d^2}{4}, \quad \sigma'_m = \sqrt{\sigma_{m,ax}^2 + 3\tau_m^2}.$$

$$\text{Goodman: } n = \left(\frac{\sigma_a}{S_e} + \frac{\sigma'_m}{S_{ut}} \right)^{-1} \text{ with } S_{ut} = 97 \text{ ksi.}^{40}$$

³⁵Shigley App. A-15 (flat-bottom grooves for retaining rings) with K_t guidance; cf. Ch. 7 Fig. 7-18 discussion of high K_t values.

³⁶Shigley App. A-15 (transverse circular hole in shaft) and Fig. 6-27 for notch sensitivity.

³⁷See Shigley shaft examples around keyseats and fillet radii (e.g., p. 390-391) and Tbl. 7-1 estimates.

³⁸Shigley Figs. 6-26 and 6-27.

³⁹Endurance-limit corrections and parameters per Shigley Eq. (6-18), Tbls. 6-2, 6-5, and Eq. (6-19).

⁴⁰Shigley Eq. (6-41) (Goodman), with combined mean components per Chs. 6-7.

Top retaining-ring groove. $M = 2039 \text{ in} \cdot \text{lbf}$, $T = 471.9 \text{ in} \cdot \text{lbf}$, $N \approx -158 \text{ lbf}$, $K_t = 4.75$,
 $K_{ts} = 3.40$, $K_f \approx 3.81$, $K_{fs} \approx 3.04$.

$$\sigma_a = 14.8 \text{ ksi}, \quad \tau_m = 1.36 \text{ ksi}, \quad \sigma_{m,ax} \approx -0.29 \text{ ksi}, \quad \sigma'_m \approx 2.38 \text{ ksi}.$$

$$n_{\text{ring}} = \left(\frac{14.8}{28.2} + \frac{2.38}{97} \right)^{-1} = \mathbf{1.82} \geq 1.5 \quad (\text{PASS}).$$

Rotor pin cross-hole section. $M = 2039 \text{ in} \cdot \text{lbf}$, $T = 471.9 \text{ in} \cdot \text{lbf}$, $N \approx -170 \text{ lbf}$, with
 $K_f = 1.92$, $K_{fs} = 2.63$, and $K_t = 2.23$. For $d = 1.750 \text{ in}$:

$$\sigma_a = \frac{32K_f M}{\pi d^3} = \boxed{7.45 \text{ ksi}}, \quad \tau_m = \frac{16K_{fs} T}{\pi d^3} = \boxed{1.18 \text{ ksi}},$$

$$A = \frac{\pi d^2}{4}, \quad \sigma_{m,ax} = K_t \frac{N}{A} = \boxed{-0.139 \text{ ksi}}, \quad \sigma'_m = \sqrt{\sigma_{m,ax}^2 + 3\tau_m^2} = \boxed{2.05 \text{ ksi}}.$$

With $S_e = 28.2 \text{ ksi}$ and $S_{ut} = 97 \text{ ksi}$,

$$n_{\text{pin hole}} = \left(\frac{7.45}{28.2} + \frac{2.05}{97} \right)^{-1} = \mathbf{3.50} \quad (\text{PASS}).$$

Pulley keyseat section. Local M is small; so torsion dominates. With $K_{ts} = 3.0$, $K_{fs} \approx 2.70$,
 $\tau_m \approx 1.22 \text{ ksi}$ and $n \gg 5$ (PASS).⁴¹

Static yield (von Mises)

With $S_y = 83 \text{ ksi}$ and the same M, T, N and K_t, K_{ts} as above,

$$\sigma_{\text{VM,max}} = \sqrt{(\sigma_b + \sigma_{m,ax})^2 + 3\tau_m^2}.$$

- Top ring groove: $\sigma_{\text{VM}} \approx 19.0 \text{ ksi} \Rightarrow n_y \approx \mathbf{4.4}$.
- Bottom ring groove (smaller M): $n_y \approx \mathbf{9.0}$.
- Pin cross-hole: $\sigma_{\text{VM}} = \sqrt{(\sigma_b + \sigma_{m,ax})^2 + 3\tau^2} \approx 8.80 \text{ ksi} \Rightarrow n_y \approx \mathbf{9.4}$.
- Keyseat at pulley (torsion only): $n_y > \mathbf{40}$.

Deflection/slope (uniform EI with $d = 1.750$)

Relative to the 1.875 in trial, stiffness scales as d^4 ; so the slopes/deflections increase by
 $\left(\frac{1.875}{1.750} \right)^4 \approx 1.31$. So the bearing slopes remain small ($\mathcal{O}(10^{-4})$ rad), still acceptable for ball
bearings.⁴²

Pulley key and set screw checks with *actual* sizes

Key: $3/8 \times 3/8$ square, $\ell = 2.625 \text{ in}$. Tangential force $F = T/r = 471.9/0.875 = 540 \text{ lbf}$. Shear:
 $\tau = \frac{F}{w\ell} = 0.55 \text{ ksi}$; bearing: $\sigma = \frac{F}{(h\ell/2)} = 1.10 \text{ ksi}$. With $S_y \approx 0.8 \cdot 97 = 77.6 \text{ ksi}$ and $n = 1.5$, both
FoS > 70 (huge margin).⁴³

⁴¹Keyseat torsion concentration estimated per Shigley shaft examples and Tbl. 7-1 order-of-magnitude guidance.

⁴²Compare to typical shaft/bearing slope considerations in Shigley Sec. 7-2 and Tbl. 7-2.

⁴³Method per Shigley Ex. 7-6.

Set screw: #10–32 316 SS on key top. Friction across the two interfaces gives $F_{\text{ax,cap}} = 2\mu N$. With $\mu = 0.15$ and design axial $F_{\text{ax,req}} = 1.5 W_{\text{pulley}} \approx 8.4 \text{ lbf}$, required $N \approx 28 \text{ lbf}$. Local bearing on the key under the screw (using $d \approx 0.190 \text{ in}$): $p \approx 1.4 \text{ ksi} \ll S_y/1.5 \approx 51.7 \text{ ksi}$.⁴⁴

Rotor taper pin and hub checks

At the pin: hub OD at the pin location is 2.559 in, so wall $t = (2.559 - 1.750)/2 = 0.4045 \text{ in}$. The size 0 pin has $D_{\text{large}} \approx 0.156 \text{ in}$ with 1:48 taper; using the previously calculated $d_1 = d_{\text{hole-large}} \approx 0.153 \text{ in}$ and $d_2 = d_{\text{hole-small}} \approx 0.117 \text{ in}$, being the diameters at the two shear⁴⁵, the double-shear area:

$$A = \frac{\pi}{4}(d_1^2 + d_2^2) = \boxed{0.0291 \text{ in}^2}.$$

Tangential force at the pulley $F = T/r = 471.9/0.875 = 540 \text{ lbf}$ gives

$$\tau_{\text{pin}} = \frac{F}{A} = \boxed{18.5 \text{ ksi}}.$$

Using the vendor single-shear breaking strength 54 ksi with design factor $n = 1.5$ ($\tau_{\text{allow}} = 36 \text{ ksi}$) yields

$$n_{\text{pin shear}} = \frac{36}{18.5} = \mathbf{1.94 \text{ (PASS)}}.$$

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Bearing/tear-out: Using the equations from the preliminary section, hub-wall bearing on the near/far walls is $p_1 \approx \boxed{4.35 \text{ ksi}}$, $p_2 \approx \boxed{5.72 \text{ ksi}}$; shaft bearing (using $d_{\text{mid}} = 0.1350 \text{ in}$) is $p_{\text{shaft}} \approx \boxed{2.29 \text{ ksi}}$; single-lug tear-out capacity $V_{\text{to}} \approx \boxed{7.78 \times 10^3 \text{ lbf}} \gg 540 \text{ lbf}$ (all PASS).⁴⁷

Taper-fit contact check The taper pin is driven into a 1:48 reamed hole; local interference δ_d at a bore of radius a produces a uniform contact pressure p along a short neighborhood that can be modeled as a solid cylinder (pin) pressed into a thick-walled cylinder (hub). The classic Lamé press/shrink-fit relation is⁴⁸

$$p = \frac{\delta_d}{a} \left[\underbrace{\frac{1 - \nu_p^2}{E_p}}_{\text{pin}} + \underbrace{\frac{1 - \nu_h^2}{E_h} \frac{b^2 + a^2}{b^2 - a^2}}_{\text{hub}} \right]^{-1}, \quad \sigma_{\theta,i} = p \frac{b^2 + a^2}{b^2 - a^2},$$

where E, ν are Young's modulus and Poisson's ratio, a is the pin/hole radius at the contact section, b is the hub outer radius, p is the radial contact pressure, and $\sigma_{\theta,i}$ is the hub hoop stress at the bore.

⁴⁴Set-screw size and properties per McMaster part 92313A832; holding-power table per Shigley Tbl. 7–4 confirms ample margin for #10 size.

⁴⁵Shigley Eq. (7–35) for small-end diameter; see also McMaster 90681A175 for ASME B18.8.2 compliance and basic dimensional tolerances.

⁴⁶Vendor “breaking strength (single shear)” per McMaster 90681A175.

⁴⁷Bearing and tear-out checks consistent with Shigley's use of $S_{sy} = 0.577S_y$ and standard lug-tear-out forms.

⁴⁸Press/shrink-fit thick-cylinder model from Shigley (interference fits; Lamé solution).

Using the actual geometry and materials at the *near wall (large end)*:

$$d_1 = 0.153 \text{ in} \Rightarrow a = \frac{d_1}{2} = 0.0765 \text{ in}, \quad b = \frac{D_{\text{hub}}}{2} = \frac{2.559 \text{ in}}{2} = 1.2795 \text{ in},$$

$$E_p = 28 \text{ Msi}, \quad \nu_p = 0.30 \quad (18-8, \text{ pin}), \quad E_h = 30 \text{ Msi}, \quad \nu_h = 0.29 \quad (\text{steel hub } S_y = 50 \text{ ksi}).$$

We have to limit p by the hub's allowable compressive/hoop stress and the pin's compressive allowable. With the project FoS $n=1.5$:

$$p_{\text{allow, hub}} = \frac{S_{y, \text{hub}}}{n} = \frac{50}{1.5} = 33.3 \text{ ksi}, \quad p_{\text{allow, pin}} \approx \frac{S_{y, 18-8}}{n} \approx \frac{30}{1.5} = 20 \text{ ksi}.$$

Solving the press-fit equation for δ_d at the near wall:

$$\delta_{d, \text{max}}^{(\text{hub})} \approx \boxed{0.000156 \text{ in}}, \quad \delta_{d, \text{max}}^{(\text{pin})} \approx \boxed{0.000094 \text{ in}}.$$

The **pin** is therefore governing - the interference cannot be greater than 0.000094in. So we can *specify a fit* with

$$\boxed{\delta_d \approx 0.00008 \text{ to } 0.00009 \text{ in at the large end}}$$

which yields (numerically):

$$p \approx 19.2 \text{ ksi}, \quad \sigma_{\theta, i} \approx 19.3 \text{ ksi},$$

both beneath their respective allowables ($\leq 20 \text{ ksi}$ for the 18–8 pin, $\leq 33.3 \text{ ksi}$ for the hub).⁴⁹ The McMaster taper pin (size 0) lists a large-end diameter tolerance of -0.0007 to $+0.0013 \text{ in}$ (ASME B18.8.2).⁵⁰ Because the hole is reamed to the same nominal taper, the effective interference is set by seating (drive) depth. To realize the target *diametral* interference δ_d (defined as $d_{\text{pin}} - d_{\text{hole}}$), specify the reamed diameters at the large and small ends as

$$d_{\text{hole, large}} = 0.15300 \text{ in} - 0.00009 \text{ in} = \boxed{0.15291 \text{ in}}, \quad d_{\text{hole, small}} = 0.11700 \text{ in} - 0.00009 \text{ in} = \boxed{0.11691 \text{ in}},$$

and control insertion depth during assembly to seat the pin within this interference band.⁵¹

Retaining ring capacity and groove bearing

Required axial with DF: $1.5 \times 145.6 = 218 \text{ lbf}$ (bottom, operating) and $1.5 \times 25.1 = 37.6 \text{ lbf}$ (top, shutdown). Vendor thrust rating $\sim 7160 \text{ lbf per ring} \Rightarrow$ huge margin. Average groove wall pressure: $p \approx \frac{F}{W\pi D_g} \approx \frac{218}{0.039 \cdot \pi \cdot 1.807} \approx 0.98 \text{ ksi} \ll 33 \text{ ksi}$ allowable on the hub (PASS).⁵²

Manufacturing Drawing and Bill of Materials

Manufacturing Notes & Tolerances

- Shaft OD as supplied: $\varnothing 1.7500_{-0.0020}^{+0.0000} \text{ in}$ (per McMaster 6117K208). Bearing fit is a free/slip fit; axial retention by external spiral rings.

⁴⁹Interference-stress form per Shigley's press/shrink-fit treatment with Lamé solution.

⁵⁰McMaster 90681A175 lists large-end diameter tolerance and ASME reference.

⁵¹McMaster 90681A175 lists large-end diameter tolerance and ASME reference.

⁵²McMaster 91663A640 lists thrust load capacity and groove geometry.

- Retaining-ring grooves (shaft): use vendor external-ring groove geometry for 1.750 in shaft; call out groove width W and for-groove diameter D_g with vendor tolerances; bottom radius r per vendor or $\approx 0.2t$. Deburr and polish groove bottom to $Ra \leq 16 \mu\text{in}$.
- Cross pin (tapered): REAM #0 (1:48) per ASME B18.8.2; seat pin to achieve effective large-end interference $\delta_d = 0.00008\text{--}0.00009$ in (contact pressure check satisfied). Show stick-out symmetry (≈ 0.125 in each side) on drawing.
- Set screw: #10-32 UNF, cup-point, Class 3A; min thread engagement $\geq d_{\text{root}}$ in pulley hub ($t_{\text{hub}} \approx 0.3125$ in).
- Key: ANSI 3/8 in square, length $\leq 1.5d$; use included key; break edges.

Manufacturing Drawing

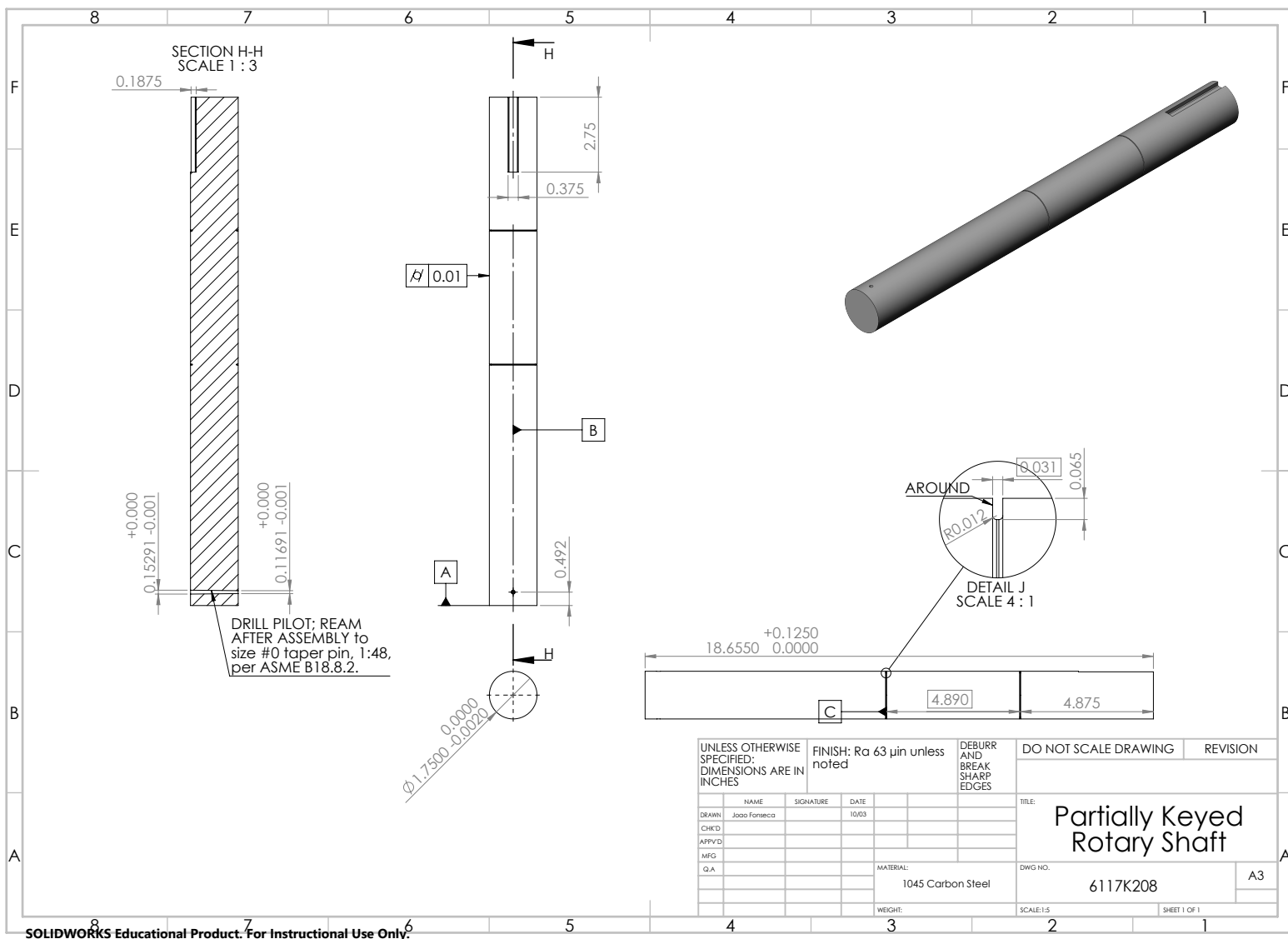


Figure 16: Manufacturing Drawing of Shaft (Pin Hole and Ring Grooves to be Machined)

Bill of Materials

Item	Description	P/N	Qty	Unit Cost
Shaft	1045 carbon steel, 1.750 in dia, partially keyed, 72 in L (will be cut to length)	6117K208	1	\$211.90 (each), \$54.90 for 18.655 in
Key	ANSI square key, 3/8 in (included with shaft)	–	1	Included
Set screw	316 SS, cup-point, #10-32, 3/4 in L (pack of 25)	92313A832	1 pack	\$9.77 / pack, \$0.39 for 1
Taper pin	18-8 SS, size 0, 2 in L (pack of 10)	90681A175	1 pack	\$15.56 / pack, \$1.56 for 1
Retaining ring	Spiral internal ring for 1.75 in ID (pack of 10)	91663A640	1 pack	\$15.04 / pack, \$3.00 for 2
<i>Estimated per-use cost (for one assembly):</i>				\$59.85

- The ANSI key is included with the specified shaft and does not require a separate purchase.

References

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3. "AISI 1050 Steel, Cold Drawn." *MatWeb*. Accessed 11 Nov. 2025. <https://www.matweb.com/>
4. "Partially Keyed Rotary Shaft, 1045 Carbon Steel, 1-3/4" Diameter, 3/8" x 3/16" Keyway, 72" Long." *McMaster-Carr*, part no. 6117K208. Accessed 11 Nov. 2025. <https://www.mcmaster.com/6117K208/>
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7. "Super-Corrosion-Resistant Cup-Point Set Screw, 316 Stainless Steel, 10-32 Thread, 3/4" Long." *McMaster-Carr*, part no. 92313A832. Accessed 11 Nov. 2025. <https://www.mcmaster.com/92313A832/>